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3D Coupled Modelling of a Series-Connected Bifilar Cu-Stabilised REBCO Resistive SFCL

Coupled H-formulation, heat-transfer and external-circuit model of a four-tape, 50 mm representative segment on a 400 V feeder

1 Introduction

System context. A directly coupled synchronous generator raises the prospective short-circuit current on a 400 V feeder above its original duty.

SFCL operation. A series resistive SFCL (Fig. 1(a)–(c)) adds negligible impedance in normal use and develops resistance in the first fault cycle [1]–[4].

Modelling requirement. Current redistribution, nonlinear E–J behaviour, Joule heating and cooling act on comparable timescales and are solved together.

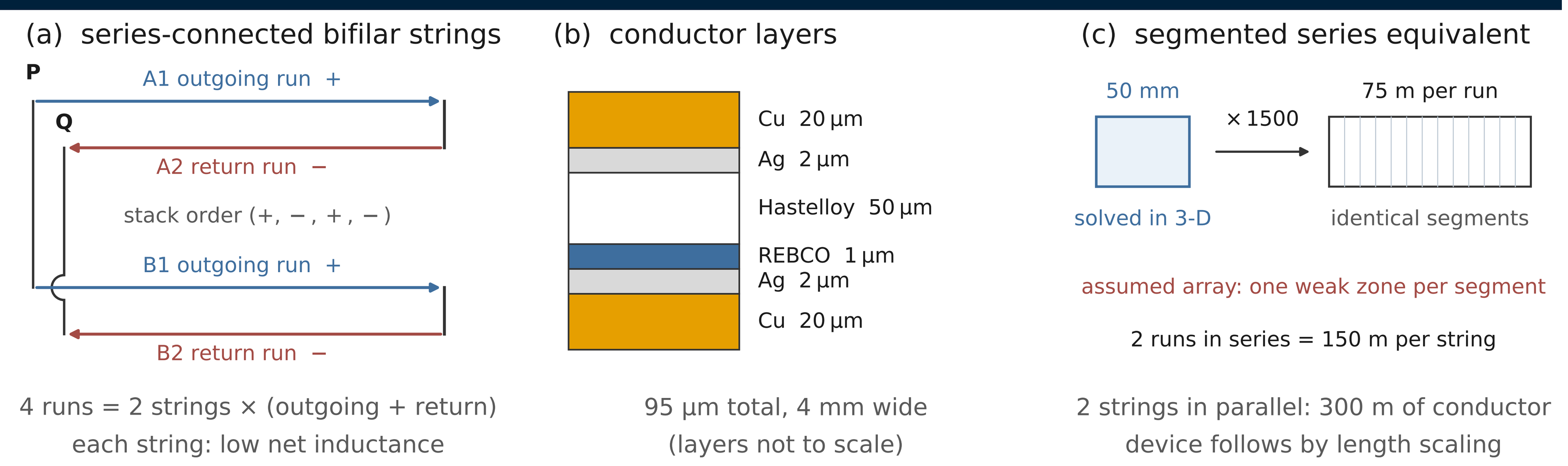


Fig. 1(a) Parallel strings of series-connected bifilar runs. **(b)** Layers. **(c)** Segment array.

2 Coupled Model and Numerical Checks

Model. The H-formulation [6] and power-law E–J relation [5] are coupled to $J_c(T, B)$, multilayer heat conduction, a 77 K boiling boundary [7] and the 400 V feeder in Figs. 1(d)–(e).

Cases. The 50 mm four-run segment uses 5 mm gaps. A 15% J_c dip is compared with an otherwise identical uniform-J_c control over clearing times of 20–80 ms.

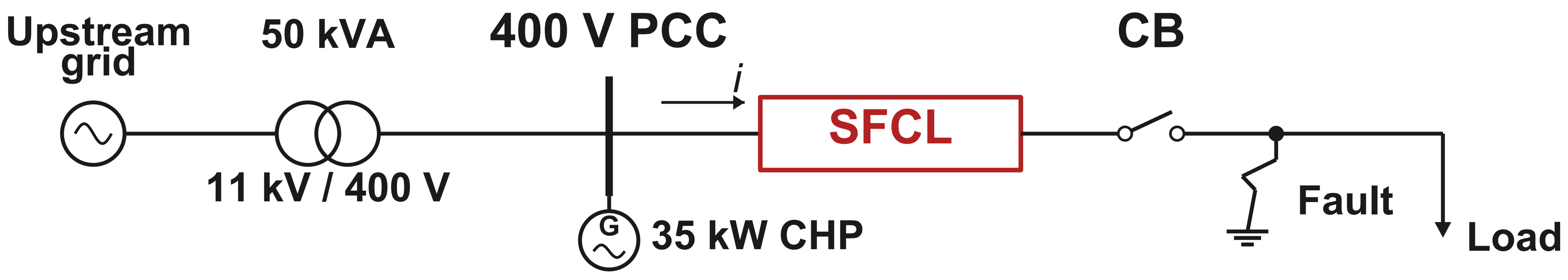


Fig. 1(d) Position of the SFCL in the 400 V grid-connected feeder.

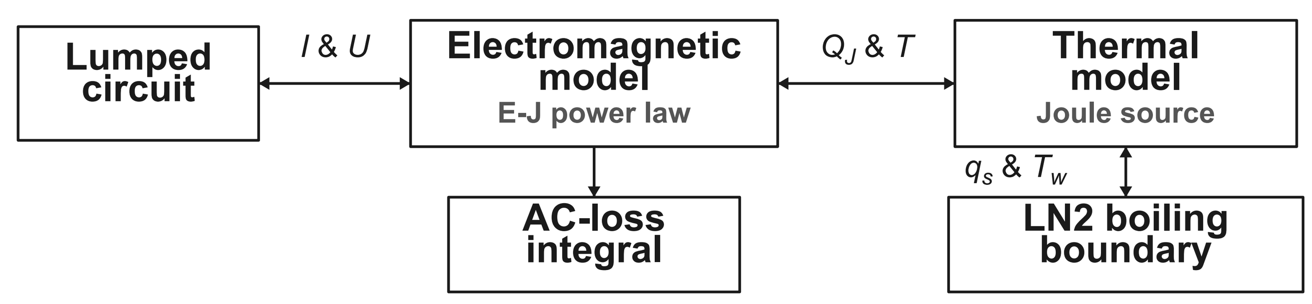


Fig. 1(e) Circuit, field, thermal and boiling coupling.

Checks. The mesh contains 119,184 elements, including one through the 1 µm REBCO thickness. The electrical-port residual is −0.81%, and the two AC-loss routes agree within 0.17%.

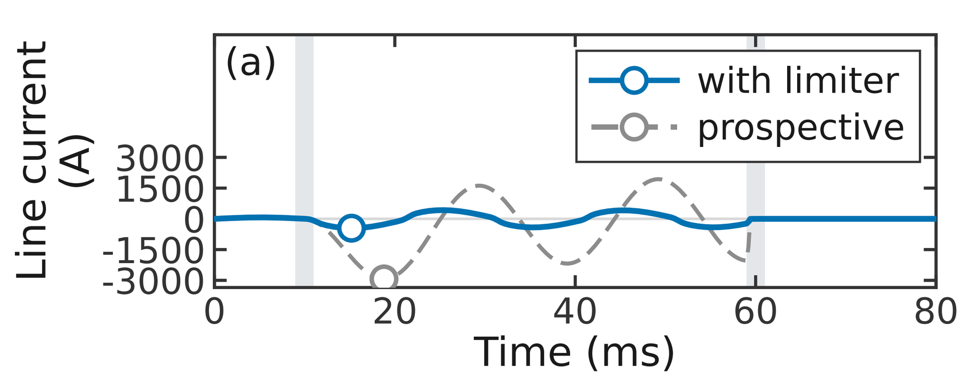
3 Results and Analysis

Current limiting. The prospective first peak of 2.93 kA is reduced to 456.1 A, a reduction of 84.4%.

Thermal response. The volume average peaks at 80.48 K and the layer maximum at 81.89 K, 8.1 K below the 90 K critical temperature.

Fault clearing. For clearing intervals of 20–80 ms, measured from fault inception to interruption, the hottest sampled point falls below 78 K within 15.2–25.0 ms after interruption.

Winding and AC loss. Counter-winding reduces the stored magnetic energy by a factor of approximately 2.8 while concentrating the magnetic field between adjacent tapes; rated-current transport loss is 5.27 W over the derived 300 m.



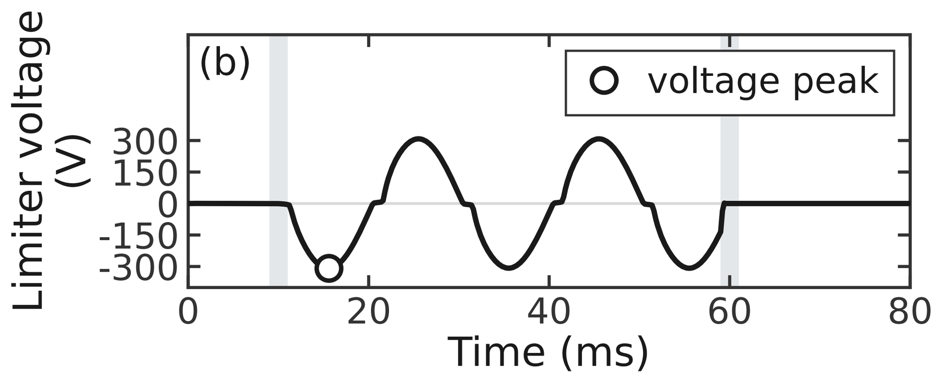


Fig. 2(a) Line current with and without the SFCL. For fault inception at 10 ms, the reduction across the four resolved half-cycle peaks is 73.7–84.4%. Dotted: uniform J_c, first peak 0.3% higher. **(b)** SFCL voltage, 308.9 V peak 0.4 ms after the current peak.

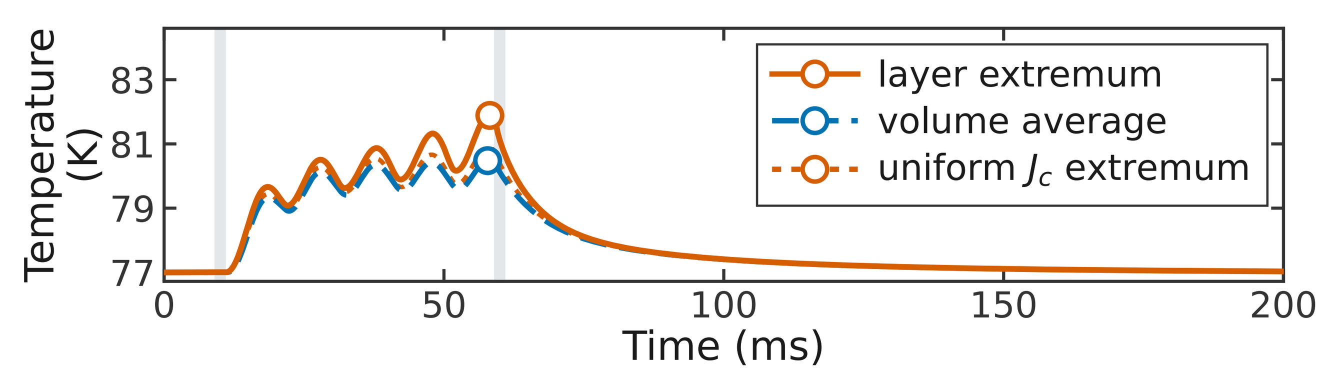


Fig. 2(c) Layer maximum and volume-average temperature;* the layer trace is the maximum over the selected REBCO surfaces. Dotted: uniform J_c, 1.0 K lower; shading: fault and breaker ramps.

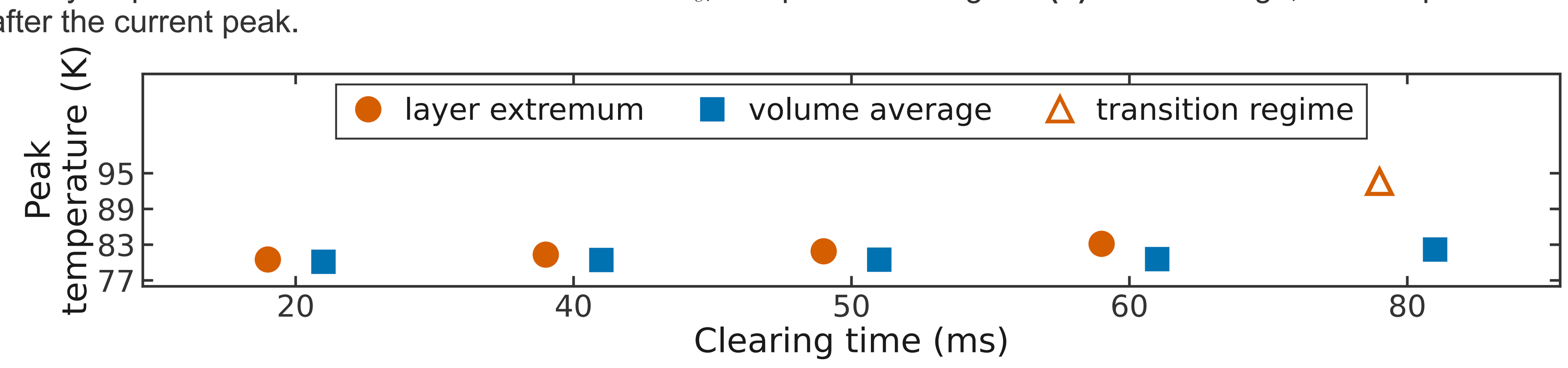


Fig. 2(d) Peak temperature against clearing time;* markers denote the five solved cases. The 80 ms case enters the prescribed boiling-transition branch, where its extremum remains mesh-sensitive.

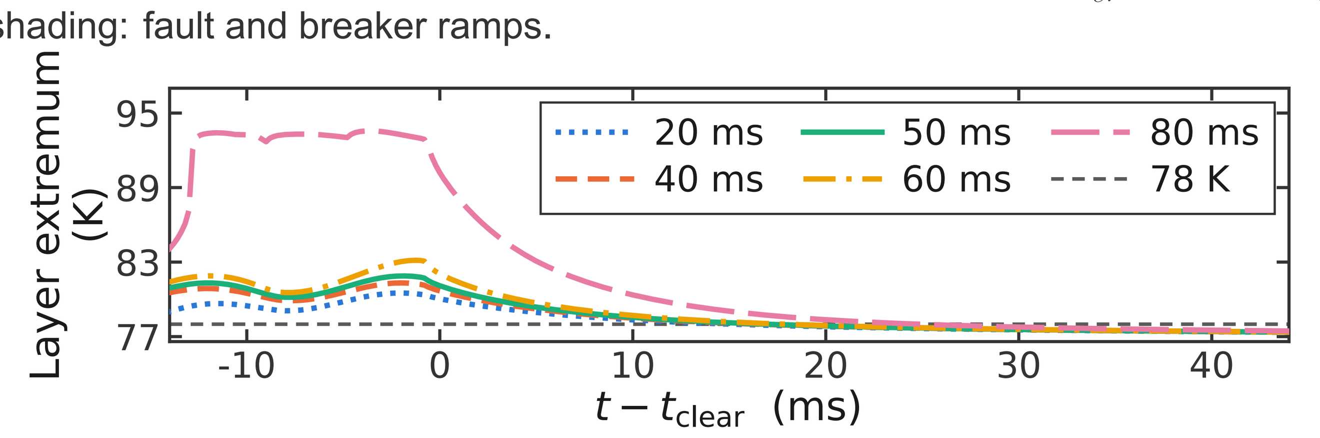


Fig. 2(e) Post-interruption cooling histories,[†] referenced to the corresponding clearing instant.

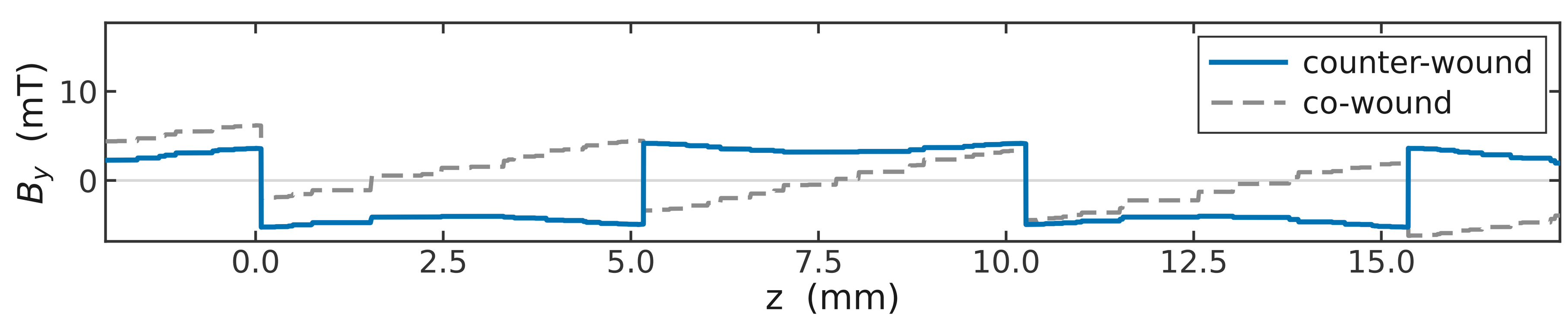


Fig. 2(f) B_y through the four-tape stack at rated current, both winding senses. The far-field share of $\int |\mathbf{B}|^2 \mathrm{d}V$ falls from 82% to 33%, the inter-tape contribution increases by 31%, and in the modelled air domain the stored energy is reduced by a factor of 2.8.

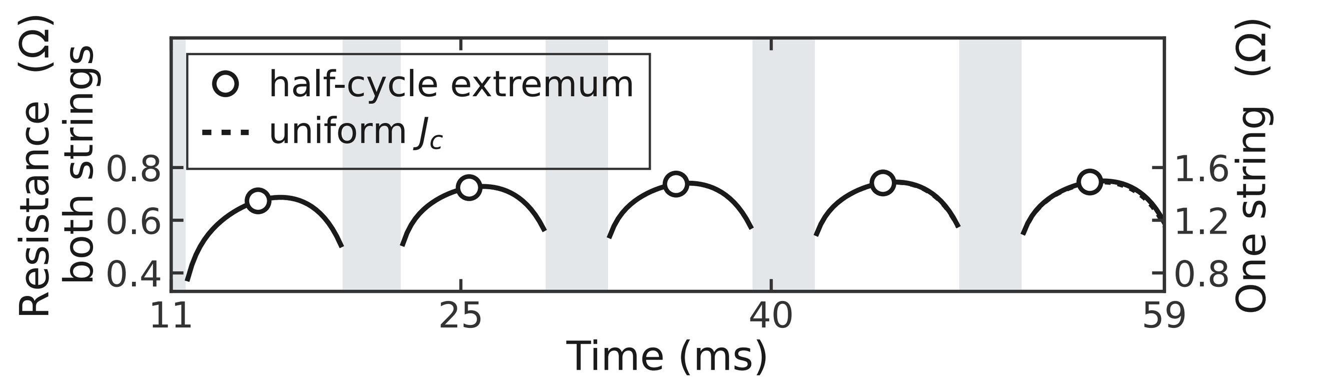


Fig. 2(g) Parallel equivalent resistance of the two strings, 0.67–0.75 Ω, evaluated for $|I| \geq 0.5\,|I|_{\mathrm{peak}}$ to suppress the zero-crossing inductive contribution. Dotted: the uniform-J_c control.

* Temperature differences below 0.6 K are not resolved by the present axial discretisation, and the 1 µm REBCO layer is represented by one element through its thickness.

† The cooling boundary uses steady, unconfined LN_2 boiling data [7], [11]. Transient heating [8] and the 5 mm gap [9], [10] may alter vapour removal and nucleate-boiling onset.

4 Conclusion

The coupled 3D model predicts an 84.4% reduction in the first fault peak on the 400 V feeder and holds the layer maximum 8 K below the critical temperature; the hottest sampled point falls below 78 K within 25 ms of every interruption computed. The 3.5 K volume-average and 4.9 K local rises characterise bulk thermal recovery and local quench development, respectively. Counter-winding reduces the stored magnetic energy by a factor of approximately 2.8 while concentrating the magnetic field between adjacent tapes.

Future work will calibrate the confined-gap LN_2 boiling boundary experimentally and assess the coupled model against device-level measurements.

References
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